

Lecture 1

1. Probability and statistics

- Probabilities of events
- Expected values
- Variance

- Summary of the data
- Estimate parameters
- Test hypothesis

2. Numerical summaries

Definitions: ① Average or mean

② Median

③ Minimum and Maximum value

④ Lower and upper quartile

e.g. Dataset: 0, 1, 4, 4, 4, ^M5, 5, 6, 6, 8, 12 ($n=11$)

Lower quartile $11 \times \frac{1}{4} = 2.75 \Rightarrow$ 第3个, $= 4$

Median 中位数位置 $\frac{n+1}{2} = 6 \Rightarrow$ Median $= 5$

Upper quartile $11 \times \frac{3}{4} = 8.25 \Rightarrow$ 第9个, $= 6$

* Q_1 (Lower): smallest data point x for which at least 25% of data $\leq x$.

Q_3 (Upper): smallest data point x for which at least 75% of data $\leq x$.

$\rightarrow$ (Interquartile Range)

IQR (四分位距) $= Q_3 - Q_1$, the distance between Q_3 & Q_1

如果 n 为偶, e.g. 1, 3, 5, 7, 9, 11 ($n=6$)

$$M = \frac{5+7}{2} = 6$$

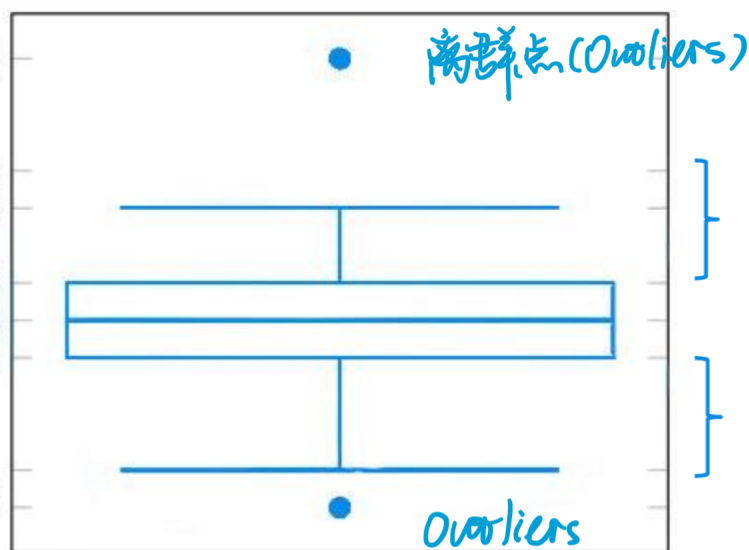
$$Q_1: 6 \times \frac{1}{4} = 1.5 \Rightarrow \text{第2个}, = 3$$

$$Q_3: 6 \times \frac{3}{4} = 4.5 \Rightarrow \text{第5个}, = 9$$

3. Boxplot (箱线图)

Data: 0 1 4 4 4 5 5 6 6 8 12

Max=12
上围栏
 $Q_3 + 1.5 \times IQR = 9$
Box, 包含一半的数
 $Q_3 = 6$
Med=5
 $Q_1 = 4$
 $Q_1 - 1.5 \times IQR = 1$
下围栏
Min=0



* This set is right-skewed, 大部分数据堆在4~6, 大值拉长分布, 反之左偏

Whisker to largest value within $1.5 \cdot IQR$

Whisker to smallest value within $1.5 \cdot IQR$

4. Graphical summaries

(1) Histograms (直方图)

- ① Divide the range of the data into intervals $B_1, \dots, B_m$ (called bins)
- ② Draw a bar of height h_i for bin B_i .

Approach 1: Set h_i equal to n_i # data points in B_i ?

(频数, n_i = 落在 B_i 里的数据个数)

Approach 2: Normalize the total area of the bars to 1

by setting h_i equal to $h_i = \frac{n_i/n}{|B_i|}$ $\rightarrow$ 该 bin 里数据比例
 $|B_i| \rightarrow$ bin 宽度

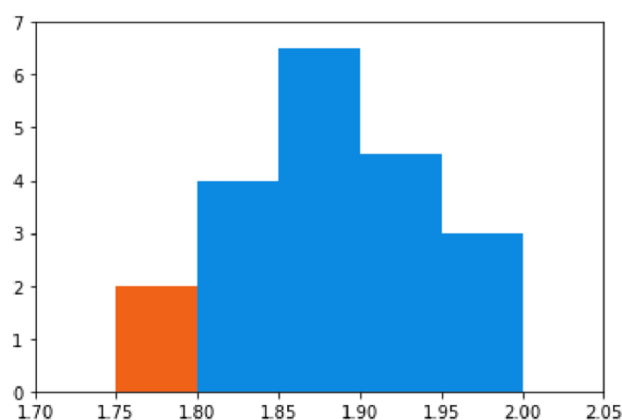
We choose approach 2 $\Rightarrow$ 每根柱的面积 = 落在该区间的数据比例.

Interpreting histograms

The histogram summarizes the height (in meters) of 40 basketball players. The histogram is normalized.

The left orange bar represents:

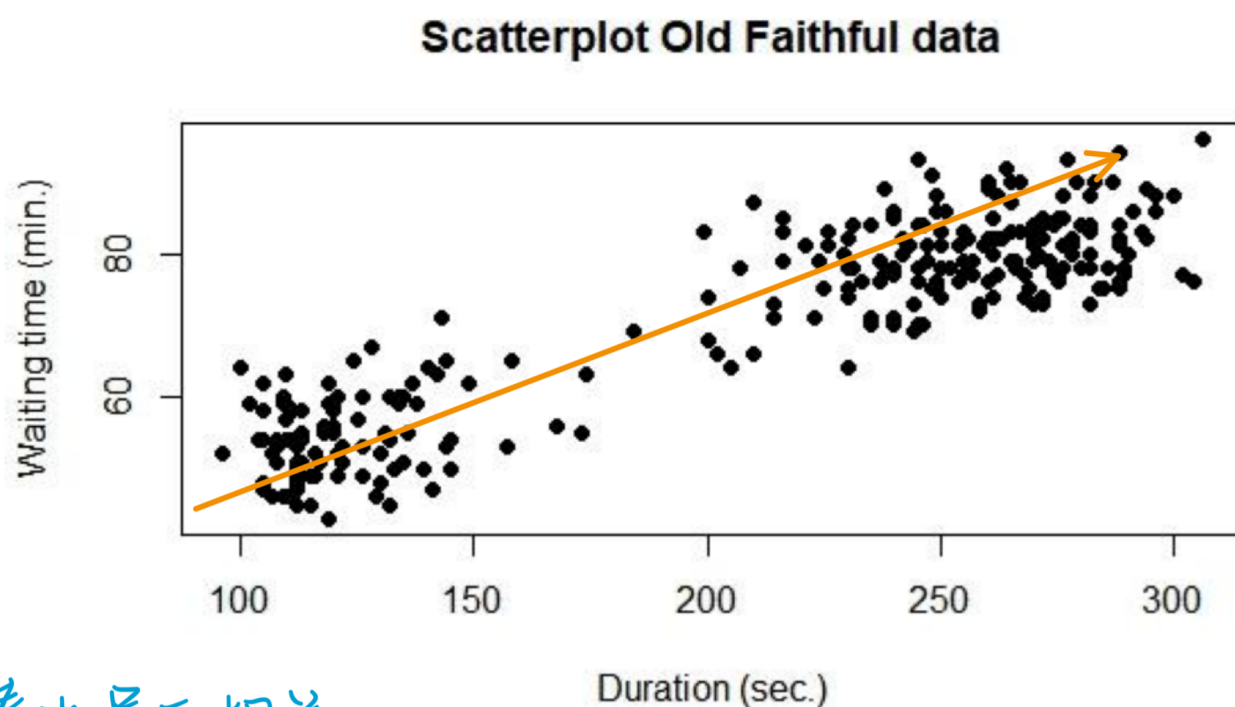
- A. 2 players
- B. 4 players**
- C. 6 players
- D. 8 players



$$2 = \frac{h_i/40}{0.05}, \quad n_i = 4$$

(2) Scatter plot: 散点图 可以同时看2个变量之间关系

e.g.



① 可以看出呈正相关.

② 分为2个 (clusters), 并且 cluster 内部仍有正趋势.

5. Sample spaces:

- ① Experiment: an activity or procedure that produces different possibilities, called outcomes.
- ② In probability and statistics, randomness is involved outcome of experiments is uncertain
- ③ The set of all outcomes is called the sample space

Notation: Sample space: $\Omega = \{\dots\}$.

Outcome is an element of the sample space: $a \in \Omega$.

e.g. We throw one die twice. The set of possible outcomes of this experiment is given by

A. $\Omega = \{2, 3, \dots, 12\}$

B. $\Omega = \{(1, 1), (1, 2), \dots, (6, 5), (6, 6)\}$
 $= \{(i, j) \mid 1 \leq i, j \leq 6\}$

C. $\Omega = \{(1, 1), (1, 2), \dots, (5, 6), (6, 6)\}$
 $= \{(i, j) \mid 1 \leq i \leq j \leq 6\}$

i 和 j 无大小关系.

D. $\Omega = \{\{1, 1\}, \{1, 2\}, \dots, \{5, 6\}, \{6, 6\}\}$ $\Rightarrow$ 花括号是 Sample space 的表示.

6. Probability functions

Definition 1: An **event** is a subset of the sample space.

Definition 2: Let Ω be a sample space. A probability function P assigns to each event A in Ω a number $P(A)$ in $[0,1]$ such that:

the probability of A

$$\left\{ \begin{array}{l} \textcircled{1}. P(\Omega) = 1 \\ \textcircled{2}. P(A \cup B) = P(A) + P(B), \text{ if } A \text{ and } B \text{ are disjoint.} \end{array} \right.$$

Lecture 2

1. Infinite sample space: Suppose we throw a die until 6 comes up. We are interested in the number of throws needed.

$$\Rightarrow \Omega = \{1, 2, 3, \dots\}$$

$$\text{Furthermore, } P(n) = \left(\frac{5}{6}\right)^{n-1} \times \frac{1}{6}$$

** A_n : 恰好第 n 次*

抛到 6

$\Rightarrow$ Is $P(n)$ a probability function?

$$\textcircled{1} P(n) \geq 0. \quad \textcircled{2} P(A_1 \cup A_2 \cup \dots) = P(A_1) + P(A_2) + \dots$$

Check: $P(\Omega) = 1$:

$$P(\Omega) = \sum_{n=1}^{\infty} \left(\frac{5}{6}\right)^{n-1} \cdot \frac{1}{6} = \frac{1}{6} \sum_{k=0}^{\infty} \left(\frac{5}{6}\right)^k = \frac{1}{6} \times \frac{1}{1 - \frac{5}{6}} = 1$$

** $|r| < 1$ 时, $\sum_{k=0}^{\infty} r^k = \frac{1}{1-r}$*

取补集 $\wedge^c$ e.g. $(E \cup F)^c = E^c \cap F^c$

2. Sum and Complement Rule

Theorem: For any two events A and B we have:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Theorem: For any event A :

$$P(A^c) = 1 - P(A)$$

3. Conditional probability

Definition: The conditional probability of A given C is:

$$P(A|C) = \frac{P(A \cap C)}{P(C)}, \text{ provided } P(C) > 0$$

Example

Two integrated circuits are tested from the same silicon wafer. A circuit is accepted (a) or rejected (r). The sample space is

$$\Omega = \{(a, a), (a, r), (r, a), (r, r)\}$$

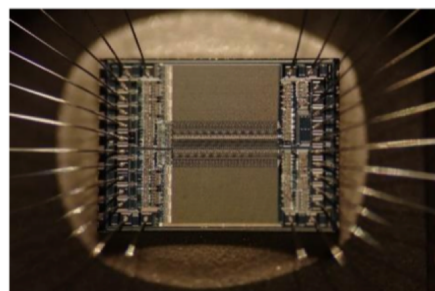
The probability model is

$$P((a, a)) = 0.95$$

$$P((a, r)) = 0.02$$

$$P((r, a)) = 0.02$$

$$P((r, r)) = 0.01$$



Determine the conditional probability that the second circuit is rejected, given that the first circuit is rejected.

$P(\text{第二片被拒} | \text{第一片被拒})$



$$P(B|A) = \frac{P(B \cap A)}{P(A)} = \frac{0.01}{0.02 + 0.01} = \frac{1}{3}$$

Multiplication rule: For any two events A and C,

$$P(A \cap C) = P(A|C)P(C)$$

* $P(A \cap B) = P(A) \cdot P(B)$ 只在 A, B 事件独立时成立

4. Bayes' rule

4) Law of total probability: motivation

$$\text{Theorem: } P(A) = P(A|C)P(C) + P(A|C^c)P(C^c)$$

Theorem: Suppose $C_1, C_2, \dots, C_m$ are disjoint events such that $C_1 \cup C_2 \cup \dots \cup C_m = \Omega$

For any event A we have:

$$P(A) = P(A|C_1)P(C_1) + P(A|C_2)P(C_2) + \dots + P(A|C_m)P(C_m)$$

e.g. Law of total probability: motivation



A company has three machines C_1, C_2 and C_3 for making resistors. The accuracy (percentage of correctly produced resistors) and production share of the machines are given in the following table:

Machine	C_1	C_2	C_3
Accuracy	80%	90%	60%
Production share	30%	50%	20%

All of the resistors are mixed together at random in one bin. What is the probability that an arbitrary resistor is accurate?

$$\begin{aligned} P(A) &= P(A|C_1) \cdot P(C_1) + \\ &P(A|C_2) \cdot P(C_2) + \\ &P(A|C_3) \cdot P(C_3) \\ &= 80\% \times 30\% + 90\% \times 50\% \\ &\quad + 60\% \times 20\% \\ &= 0.81 \end{aligned}$$